Take-Home Part II

Sam Olson

Question 1

A major difficulty in what we did previously was determining a good covariance matrix to use in the construction of confidence intervals based on asymptotic normality, regardless of whether that asymptotic normality was based on maximum likelihood or maximum composite likelihood estimation. One possible alternative to the direct use of asymptotic results is to compute intervals directly using parametric bootstrap methods.

(a)

> Determine 95% confidence intervals for the five parameters of the model through the application of percentile bootstrap intervals for maximum likelihood estimation.

The 95% percentile bootstrap confidence intervals for all five parameters are reported in Table 1; the construction is described below.

We construct 95% percentile bootstrap confidence intervals for each of the five model parameters $(\alpha, \beta, \gamma, \theta, \sigma^2)$ by following expression (10.15), where the $(1-\tilde{\alpha})100\%$ percentile bootstrap interval for a scalar parameter is given by the empirical $\tilde{\alpha}/2$ and $1-\tilde{\alpha}/2$ quantiles of the M bootstrap estimates. The use of $\tilde{\alpha}$ is to distinguish between the parameter of interest and the "typical" usage for quantiles.

We take $M = 999$ so that $(M+1) \cdot 0.025 = 25$ is an integer, satisfying the integrality condition noted after expression (10.10). For each bootstrap replicate $m = 1, \ldots, M$, we simulate $\mathbf{y}_m^* = (y_1^*, \ldots, y_n^*)^T$ from the fitted model $F(\mathbf{y} \mid \hat{\theta}_n)$. However, because the model includes AR(1) errors, simulation requires drawing $\varepsilon_1^* \sim N\Big(0,\, \hat{\sigma}^2/(1-\hat{\theta}^2)\Big)$ from the stationary distribution and then propagating $\varepsilon_i^* = \hat{\theta}\,\varepsilon_{i-1}^* + \delta_i$ with $\delta_i \overset{iid}{\sim} N(0, \hat{\sigma}^2)$ for $i = 2, \ldots, n$. The simulated response is then $y_i^* = \hat{\mu}_i + \varepsilon_i^*$ where $\hat{\mu}_i = \hat{\alpha}\, x_i^{\hat{\beta} x_i^{-\hat{\gamma}}}$.

Each bootstrap dataset $\mathbf{y}_m^*$ is re-fit by maximizing the same log-likelihood as in Q3 from Part I, using the original MLE $\hat{\theta}_n$ as the starting point for BFGS optimization to get our bootstrap estimates.

Regarding these estimates: The parametric bootstrap approximates the sampling distribution of the estimator as a *procedure*, not solely as a functional of the data. As such, the estimator is defined implicitly through a numerical optimization routine, which may fail to converge for certain bootstrap datasets; such replicates are discarded, but not resampled so as to replace those bootstrap estimates. Doing so would alter the bootstrap data-generating mechanism by preferentially retaining datasets for which the likelihood surface is sufficiently regular, potentially excluding replicates corresponding to extreme parameter estimates and biasing the empirical distribution toward regions of the sample space where estimation is numerically stable.

That said, of the $M = 999$ replicates, $M_{\text{eff}} = 988$ converged successfully and were retained (using `set.seed(42)`). Inference is therefore based on the empirical distribution of $\hat{\theta}_{n,m}^*$ (our boostrap estimates) conditional on successful convergence of the re-estimation procedure applied to $\mathbf{y}_m^*$. Following expression (10.15), the 95% percentile interval for each parameter is the empirical 2.5% and 97.5% quantiles of the $M_{\text{eff}} = 988$ bootstrap estimates, computed via interpolation rather than exact order-statistic indexing

since $(M_{\text{eff}} + 1) \cdot 0.025 = 24.725$, which is not an integer. This conditioning may result in underestimating uncertainty, but with only 11 failures out of 999 the effect is expected to be negligible. Also, no discrepancy function $h(\theta, \hat{\theta}_n)$ needs to be formed or inverted because the percentile method works directly on the empirical distribution of $\theta^*_{n,m}$.

The asymptotic-normal intervals from Q3 are included in Table 1 for comparison. The concerns of Part I regarding whether the relevant asymptotics hold are not eliminated by this approach, but the bootstrap avoids the need to invert the observed information matrix directly.

Table 1: 95% Percentile Bootstrap and Asymptotic-Normal CIs

Parameter	MLE	Boot 2.5%	Boot 97.5%	AN 2.5%	AN 97.5%
α	0.7395	0.0675	2.5107	-0.4834	1.9625
β	1.0283	0.3085	2.5010	0.0275	2.0291
γ	0.1574	0.0552	0.2099	0.0929	0.2219
θ	0.5598	-0.1289	0.6488	0.2203	0.8993
σ^2	0.6250	0.2633	0.8790	0.2785	0.9714

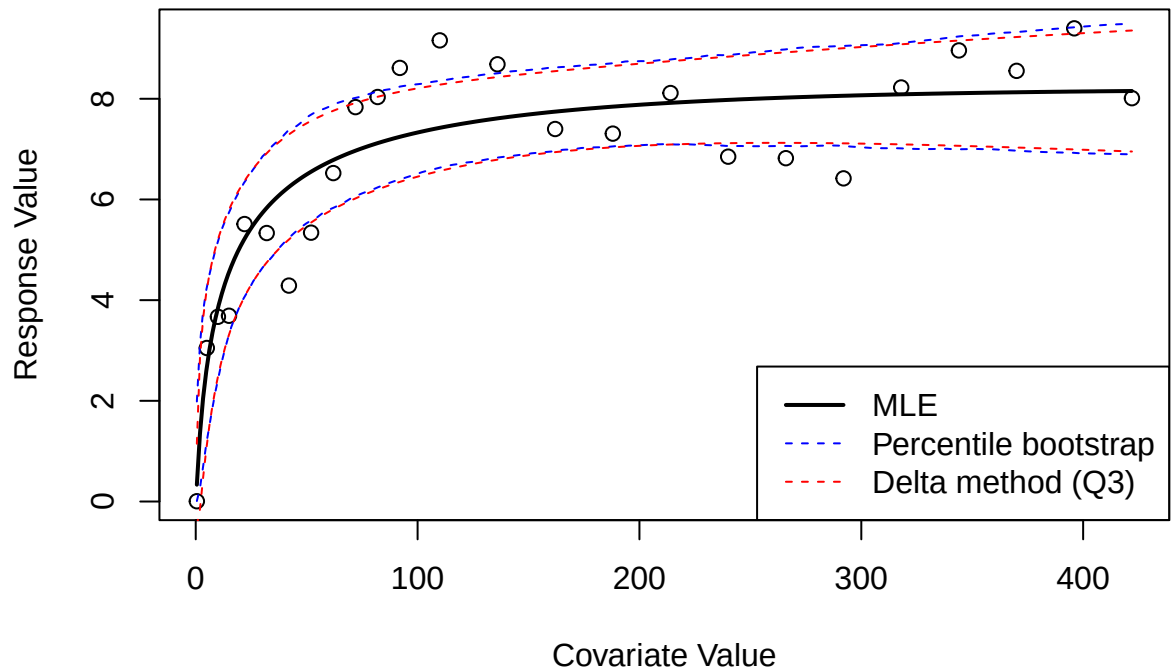
(b)

> Determine a 95% confidence band for the expectation function, also using percentile bootstrap methodology with maximum likelihood estimation.

We construct a *pointwise* 95% confidence band for the expectation function $\mu(x; \alpha, \beta, \gamma) = \alpha x^{\beta x^{-\gamma}}$, meaning the nominal 95% coverage applies at each fixed x_0 separately, not simultaneously over the full curve. The figure below shows the MLE-centered fit, the percentile bootstrap band, and the delta-method band (described below), all evaluated on an equally spaced grid of x-values spanning the observed covariate range.

For each retained bootstrap replicate $m = 1, \ldots, M_{\text{eff}}$, the re-estimated parameters $(\hat{\alpha}^*_m, \hat{\beta}^*_m, \hat{\gamma}^*_m)$ from part (a) are used to evaluate the fitted curve over the aforementioned grid, yielding $\mu^*_m(x_0) = \hat{\alpha}^*_m \, x_0^{\hat{\beta}^*_m x_0^{-\hat{\gamma}^*_m}}$ at each grid point x_0. Following expression (10.15), the pointwise 95% percentile interval at x_0 is the empirical 2.5% and 97.5% quantiles of $\{\mu^*_m(x_0) : m = 1, \ldots, M_{\text{eff}}\}$, which required interpolation as $(M_{\text{eff}}+1) \cdot 0.025 = 24.725$ is not an integer.

For comparison, a delta-method band is also shown, though wasn't explicitly asked for in this section. That said, both bands are centered on the same MLE curve, such that differences reflect methodology in the interval construction. Though commented in greater detail later, it is interesting that while both the bootstrap and delta-method intervals look fairly similar, the bootstrap band is asymmetric and wider in some regions, while the delta-method band is symmetric and narrower throughout.



2.

Another possibility that we have at our disposal is to approach the problem through the use of Bayesian methods. Fit the model with extended Freundlich expectation function and autoregressive error structure using Bayesian analysis. To make this a manageable assignment, I will give some additional guidance. First, make use of the following prior distributions:

- For α, use a normal prior with mean 0.50 and variance 10, but truncate this prior at 0 to enforce the parameter space.
- For β, use a gamma prior with parameters 0.01 and 0.01 parameterized to give an expected value of 1 and a variance of 100.
- For γ, use an exponential prior with parameter 15. Note that this prior is quite informative and places probability 0.98 below a value of 0.25. This prior is chosen based on the rather extreme effect of the parameter γ on the response function.
- For θ, use a Laplace (or double exponential) prior with parameter 0.35, truncated below at -1 and above at 1. The value of 0.35 in a nontruncated Laplace distribution results in probability of 0.95 between -1 and 1, so the truncation is not severe, but we might want to preclude (eliminate) the possibility of a non-stationary error process for the w_i.
- For σ (not σ^2), use a uniform prior on the interval $(0, 20)$.

Use an overall Gibbs algorithm with several Metropolis pieces, one for the parameters α and β together, one for the parameter γ by itself, and the third for the parameters θ and σ^2 together. Note that, regardless of whether you express the likelihood and posterior in terms of σ^2 or σ, the uniform prior is assigned to σ.

As part of your answer, you should execute and report on the following:

(a)

> Using the several short chains approach, determine an appropriate burn-in period based on an examination of the parameter α. You do not need to compute the Gelman-Rubin scale reduction factor, but you should present a trace plot with 3 chains started at different (as dispersed as you can reasonably manage) starting values. Also present a plot of autocorrelations for values of α. Note that in an actual problem of this type we might want to do these things for each of the parameters, but for this exam we will restrict attention to α.

Analytical justification

Given my question is class regarding truncation of distributions, I wanted to be explicit about the methodology I used here.

To start, our target posterior in this problem is proportional to the product of the likelihood and the prior:

$$p(\alpha, \beta, \gamma, \theta, \sigma \mid \mathbf{y}) \;\propto\; L(\alpha, \beta, \gamma, \theta, \sigma^2 \mid \mathbf{y}) \cdot \pi(\alpha) \cdot \pi(\beta) \cdot \pi(\gamma) \cdot \pi(\theta) \cdot \pi(\sigma)$$

The likelihood with AR(1) errors $\varepsilon_i = w_i - \theta w_{i-1}$, $w_i = y_i - \mu(x_i; \alpha, \beta, \gamma)$, is given by

$$L \;=\; (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n} \varepsilon_i^2 \right)$$

This is the full conditional likelihood, treating $w_0 = 0$; the exact form would additionally include the marginal $w_1 \sim N(0, \sigma^2/(1-\theta^2))$.

We then need to specify the densities of the parameters involved. Starting with α, the untruncated $N(0.5, 10)$ density is $\phi\left(\frac{\alpha - 0.5}{\sqrt{10}}\right)/\sqrt{10}$. Truncating to $(0, \infty)$ renormalizes by the probability mass above zero:

$$\pi(\alpha) = \frac{\phi\left(\frac{\alpha-0.5}{\sqrt{10}}\right)/\sqrt{10}}{1 - \Phi\left(\frac{0-0.5}{\sqrt{10}}\right)} \cdot \mathbf{1}(\alpha > 0)$$

In the MH ratio this normalizing constant cancels when both current and proposed values satisfy $\alpha > 0$, which is why returning $-\infty$ for $\alpha \le 0$ is sufficient and no explicit constant is needed in the code.

For θ, the Laplace$(0, 0.35)$ density is $\frac{1}{2(0.35)} \exp(-|\theta|/0.35)$. Truncating to $(-1, 1)$ gives

$$\pi(\theta) = \frac{\exp(-|\theta|/0.35)}{2(0.35)\left(1 - e^{-1/0.35}\right)} \cdot \mathbf{1}(|\theta| < 1)$$

Again the normalizing constant cancels in the MH ratio, so only $-|\theta|/0.35$ enters `log_prior`, with hard rejection for $|\theta| \ge 1$.

Up to this point, the full conditional likelihood was considered; however, we'll take logs to ensure computational stability. This means the unnormalized log-posterior is of the form

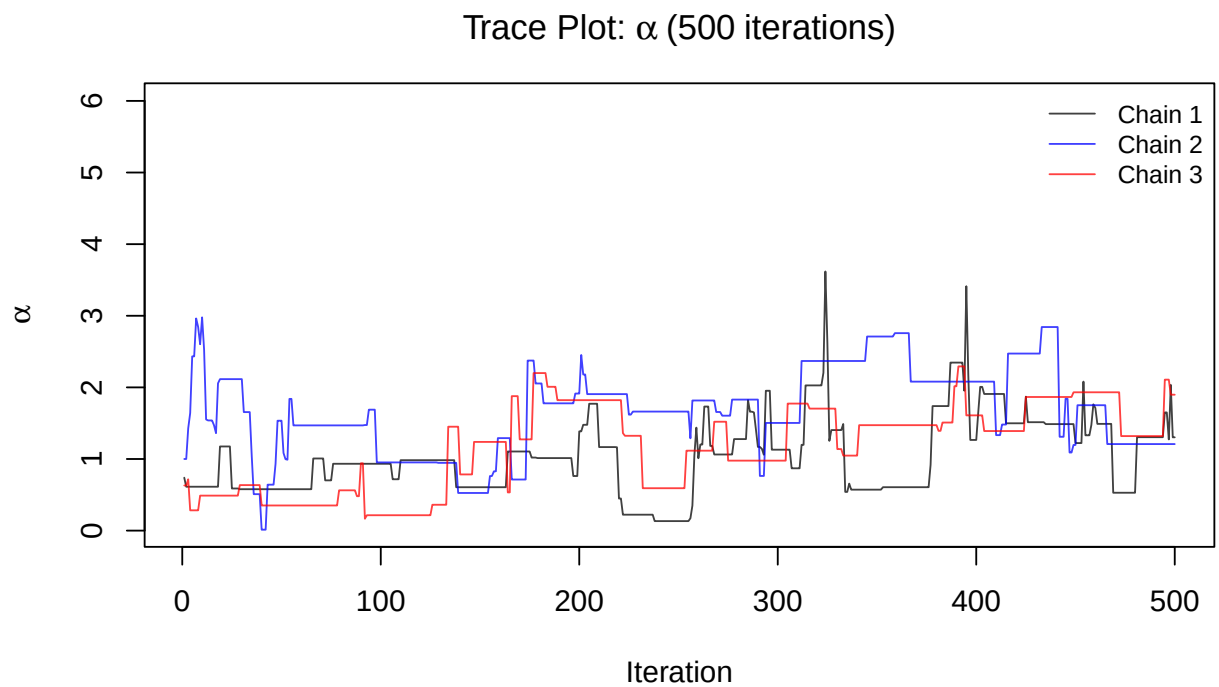
$$\log p(\boldsymbol{\psi} \mid \mathbf{y}) = \log L + \log \pi(\alpha) + \log \pi(\beta) + \log \pi(\gamma) + \log \pi(\theta) + \log \pi(\sigma)$$

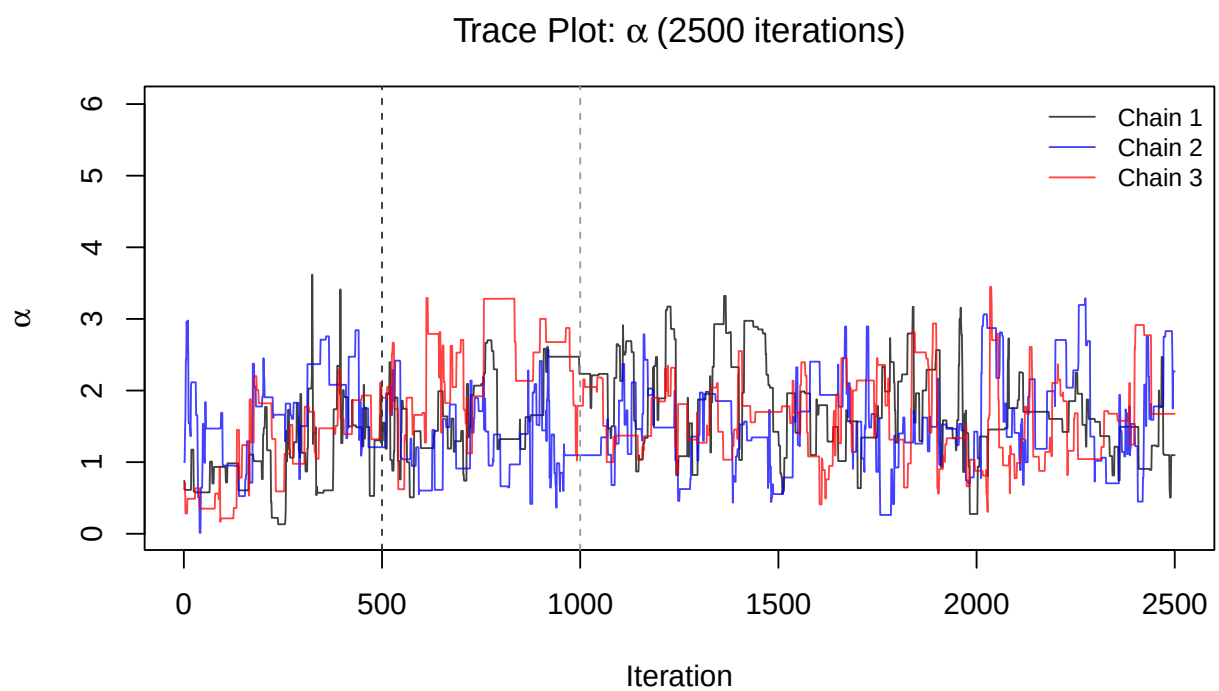
with

$$\ell = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_i \varepsilon_i^2.$$

At each Metropolis step a proposal $\boldsymbol{\psi}^*$ is accepted with probability $\min\left(1,\ e^{\log p(\boldsymbol{\psi}^*|\mathbf{y}) - \log p(\boldsymbol{\psi}^{(t)}|\mathbf{y})}\right)$, evaluated in code as `log(runif(1)) < lp_proposed - lp_current`.

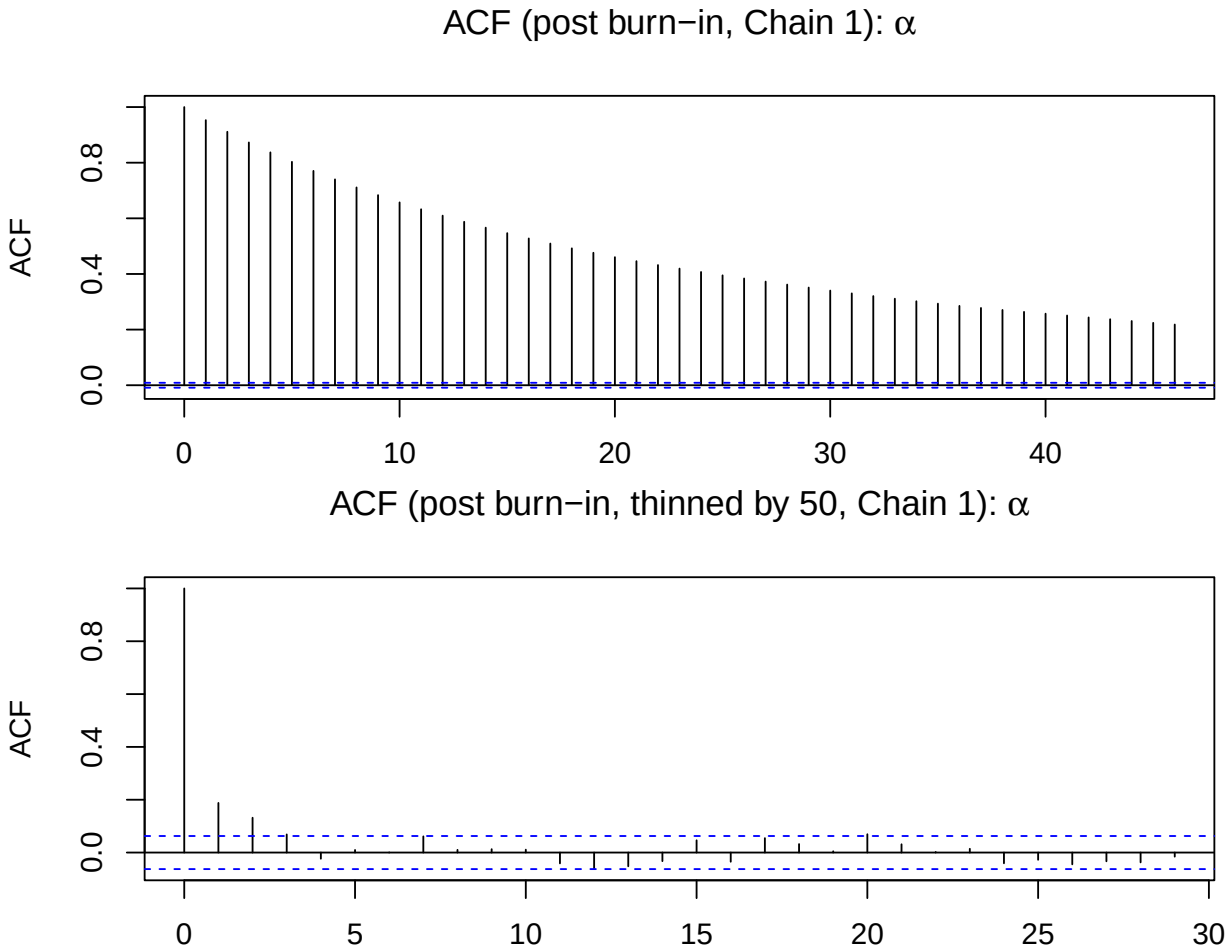
Computation





A burn-in of 1,000 iterations is selected. Three chains were initialized from (relatively) dispersed starting values and run for several short trial lengths. By 500 iterations the chains are traversing overlapping ranges with no systematic drift or between-chain separation, indicating that transient dependence on the

starting values has dissipated. Behavior is visually indistinguishable from the remainder of the 2,500-iteration run. The 500-iteration mark is taken as the apparent convergence threshold; we then double it.



The raw post-burn-in ACF for α (top) reveals substantial autocorrelation remaining well above significance bounds even at lag 50, indicating slow mixing. Two sources contribute to this. First, although α and β are updated jointly in the same MH block, an random walk proposal still struggles to efficiently traverse a strongly correlated posterior surface within that block. Second, and more fundamentally, γ is updated in a separate MH block yet is itself highly correlated with α and β (Table 2); this between-block correlation is never jointly addressed by the sampling structure, imposing an additional bottleneck on mixing. Thinning by a factor of 50 (bottom) reduces autocorrelation to negligible levels beyond lag 1, confirming that draws separated by 50 iterations are effectively independent.

Thinning is used here diagnostically to characterize mixing, not as a required step for inference. The practical consequence is that the effective sample size is considerably smaller than the nominal $49,000$ retained draws, and the several short chains approach, while appropriate for burn-in diagnosis via between-chain behavior, would yield effectively negligible inferential value at chain lengths adequate for convergence assessment alone. I believe that is why the posterior summaries in part (c) are drawn from a single long chain of $50,000$ iterations with the first 1000 discarded.

Table 2: Estimated correlation matrix of MLEs

	alpha	beta	gamma
alpha	1.000	-0.968	-0.685
beta	-0.968	1.000	0.843
gamma	-0.685	0.843	1.000

(b)

The proposal densities should all be random walks, and you will need to tune the Metropolis portions of your overall algorithm. Begin with a random walk variance of 0.50 for all parameters and tune the algorithm through trial and error to hopefully achieve acceptance rates of somewhere in the range of about 0.20 to 0.50. Present a table that shows variance specifications and acceptance rates. You do not need to show every step you used in your trial and error process, but show at least the beginning, the end, and one or maybe two places in between.

Table 3: Metropolis tuning: proposal SDs and acceptance rates.

						Acceptance Rate		
Step	σ_α	σ_β	σ_γ	σ_θ	σ_σ	Block 1	Block 2	Block 3
Initial	0.50	0.50	0.50	0.50	0.50	0.068	0.029	0.132
Attempt 2	0.80	0.80	0.05	0.50	0.50	0.041	0.236	0.132
Attempt 3	0.30	0.30	0.03	0.35	0.35	0.098	0.310	0.203
Final	0.10	0.10	0.03	0.25	0.25	0.297	0.267	0.335

Note:
Block 1: (α, β) jointly. Block 2: γ. Block 3: (θ, σ) jointly.

Table 3 reports the initial, two intermediate, and final tuning configurations. All tuning is conducted on the standard deviation scale; corresponding variances appear in the table footnote. Acceptance rates at each stage were averaged across all three chains over 2,000 iterations to avoid results being an artefact of any single run.

For the initial attempt, Block 2 (γ) had an acceptance rate near 0.029, far below target, because the informative Exp(15) prior concentrates γ tightly near zero, making large proposals almost always land in low-probability territory. The γ SD was reduced to 0.05. Block 1 (α, β) and Block 3 (θ, σ) were both below target, so their SDs were increased. This overcorrected Block 1 (0.041), so in Attempt 3 the SDs were pulled back to 0.3 and 0.35 respectively, with γ refined to 0.03. Block 1 remained below target (0.098), motivating the final reduction.

The final proposal specifications achieve acceptance rates of 0.297, 0.267, and 0.335 for Blocks 1, 2, and 3 respectively, all within the target range of 0.20–0.50.

(c)

Collect $M = 50{,}000$ samples from the joint posterior after burn-in. Summarize the results using five number summaries, 95% credible intervals, and correlations in the joint posterior. Compare the point estimates and confidence intervals to what was obtained previously for maximum likelihood using a (probably incorrect) information matrix and using parametric bootstrap.

Overall, the discrepancies between methods are largest where priors are more informative or the asymptotically normal intervals cross parameter boundaries (α, θ), and smallest where the likelihood dominates (γ). The asymptotically normal intervals from Part I are least trustworthy throughout, most visibly for α, where both the bootstrap and Bayesian intervals offer clear improvement. Moreover, the strong posterior correlations of Table 6 suggest substantial parameter trading-off in the likelihood, which tends to produce a poorly-invertible observed information matrix (singular or nearly singular) and unreliable asymptotically normal intervals across all five parameters, perhaps an unsurprising conclusion given the results of Part I.

Specifically, for each parameter:

The differences across methods are most striking for α: The posterior median (1.692) is more than double the MLE (0.740), and the Bayesian credible interval (0.309, 3.327) is shifted substantially rightward relative to both the bootstrap (0.068, 2.511) and asymptotically normal (AN) intervals. The AN interval is invalid, including negative values and violating the parameter constraint $\alpha > 0$. This is the clearest illustration of why the information matrix was described as "probably incorrect".

For β, the posterior median (0.474) is roughly half the MLE (1.028). The Bayesian interval (0.181, 1.574) is notably narrower and lower than the bootstrap (0.309, 2.501). Due to strong posterior correlation ($\rho_{\alpha\beta} = -0.798$), combinations of higher α and lower β yield similar fitted values; and the marginal MLE captures neither the shift nor the tradeoff.

For γ, the relatively informative Exp(15) prior is visibly pulling the posterior median (0.101) below the MLE (0.157), and the Bayesian interval (0.008, 0.202) is narrower than the bootstrap (0.055, 0.210). This is perhaps to be expected (by design) as the prior was explicitly chosen to constrain γ based on subject-matter knowledge about its extreme effect on the response function.

For θ, the posterior median (0.620) is modestly above the MLE (0.560), and the Bayesian interval (0.220, 0.957) sits entirely in positive territory. The bootstrap interval (-0.129, 0.649) includes negative values, suggesting the Laplace prior with its concentration near zero is providing meaningful regularization away from the lower tail.

For σ^2, the posterior median (0.868) is notably higher than the MLE (0.625), and the Bayesian credible interval (0.481, 1.714) is considerably wider than both the bootstrap (0.263, 0.879) and AN intervals (0.279, 0.971). This makes sense for two reasons. First, the Bayesian analysis propagates uncertainty from all other parameters into the marginal posterior for σ^2, whereas the MLE treats the other parameters as fixed at their point estimates. Second, the uniform prior on σ implies a non-uniform induced prior on σ^2 proportional to $1/(2\sqrt{\sigma^2})$, placing relatively more prior mass in the upper tail of σ^2 and pulling the posterior upward relative to the MLE.

Table 4: Five-number summary of posterior samples ($M = 50{,}000$).

Parameter	Min	Q1	Median	Q3	Max
α	0.0339	1.1537	1.6923	2.3007	4.5775
β	0.0012	0.3348	0.4743	0.6869	2.0691
γ	0.0000	0.0611	0.1007	0.1434	0.4249
θ	-0.1373	0.4854	0.6204	0.7551	0.9999
σ^2	0.2937	0.6982	0.8684	1.0897	4.5113

Table 5: Comparison of 95% intervals: Bayesian credible, percentile bootstrap, and asymptotic normal.

			Bayes		Bootstrap		Asym. Normal	
Parameter	Post. Med.	MLE	2.5%	97.5%	2.5%	97.5%	2.5%	97.5%
α	1.6923	0.7395	0.3088	3.3273	0.0675	2.5107	-0.4834	1.9625
β	0.4743	1.0283	0.1811	1.5737	0.3085	2.5010	0.0275	2.0291
γ	0.1007	0.1574	0.0082	0.2024	0.0552	0.2099	0.0929	0.2219
θ	0.6204	0.5598	0.2196	0.9572	-0.1289	0.6488	0.2203	0.8993
σ^2	0.8684	0.6250	0.4808	1.7139	0.2633	0.8790	0.2785	0.9714

Table 6: Posterior correlations among parameters ($M = 50{,}000$ samples).

	α	β	γ	θ	σ^2
α	1.0000	-0.7977	-0.4599	0.0300	0.2474
β	-0.7977	1.0000	0.7507	-0.0344	-0.2265
γ	-0.4599	0.7507	1.0000	0.0551	-0.2142
θ	0.0300	-0.0344	0.0551	1.0000	0.0143
σ^2	0.2474	-0.2265	-0.2142	0.0143	1.0000

(d)

A scatterplot of the data with the median expectation function and a pointwise 95% credible band. Why might we prefer using the posterior medians here rather than the posterior means? Compare this Bayesian credible band to those constructed using likelihood theory, both asymptotic normality and as constructed by parametric bootstrap.

Regarding Posterior medians:

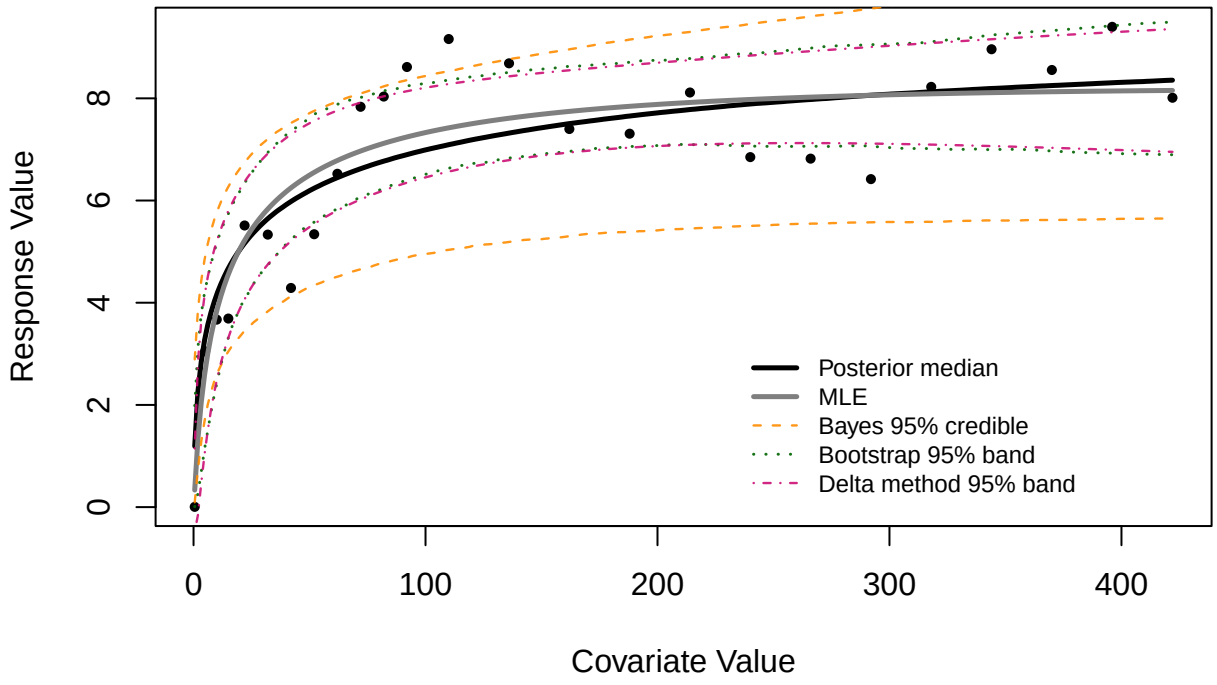
Posterior medians are preferred here for two related reasons. First, they are less sensitive to skewness in the marginal posterior distributions. This is relevant in this instance because the posteriors of α and β are right-skewed, evidenced in the five-number summaries where Q3 − Median > Median − Q1 for both parameters; in addition, α has a hard lower boundary at zero imposed by the truncated normal prior. By contrast, the posterior mean is more sensitive to skewness, making it a less stable summary of central tendency compared to the median.

More fundamentally, for a nonlinear function like the extended Freundlich, plugging in marginal means is problematic on two counts: The marginal means may not correspond to any coherent parameter vector given the strong posterior correlations documented in Table 6, and the nonlinearity of μ means that $E[\mu(x;\boldsymbol{\psi})] \neq \mu(x; E[\boldsymbol{\psi}])$ in general, via Jensen's. Note that the plug-in curve $\mu(x;\tilde{\alpha},\tilde{\beta},\tilde{\gamma})$ evaluated at the posterior medians is not the same as the pointwise posterior median of $\mu(x)$ computed directly from the Monte Carlo draws; the latter is what is reported here, and is the preferred summary for both reasons given.

Comparing bands:

The most immediately striking feature of the below plot is the separation between the MLE and the posterior median: The posterior median curve sits visibly above the MLE throughout the middle of the covariate range, a direct consequence of the upward shift in the marginal posterior of α relative to its MLE, noted in part (c). Regarding the bands themselves, the Bayesian credible band is the widest of the three at virtually every covariate value, and notably so at high covariate values where data are sparse, reflecting joint uncertainty across all five parameters via the posterior distribution. The bootstrap slightly different from the AN band, and the bootstrap band shares the Bayesian band's asymmetry, since both respect the nonlinearity of the expectation function rather than relying on a linear approximation.

In summary: The Bayesian band is widest and asymmetric, accounting for full parameter uncertainty; the bootstrap band is intermediate (compared to the other too) while also being asymmetric, meaning both methods respect the nonlinearity of μ without a linear approximation. By contrast, the delta-method band from Part I is narrowest and symmetric, centered on the same MLE curve as the bootstrap band but underestimating uncertainty, most visibly at large covariate values where data are sparse.



Appendix - Gibbs Sampler

```r
# Log-likelihood under AR(1) error structure
log_lik <- function(alpha, beta, gamma, theta, sigma2, x, y, n) {
  mu <- alpha * x^(beta * x^(-gamma))
  w <- y - mu
  eps <- w - theta * c(0, w[-n])
  -(n/2)*log(2*pi) - (n/2)*log(sigma2) - sum(eps^2)/(2*sigma2)
}

# Log prior - unnormalized, truncation constants cancel in MH ratio
log_prior <- function(alpha, beta, gamma, theta, sigma) {
  if (alpha <= 0 || beta <= 0 || gamma <= 0 ||
      sigma <= 0 || sigma > 20 || abs(theta) >= 1) return(-Inf)
  lpa <- dnorm(alpha, 0.5, sqrt(10), log = TRUE) # N(0.5, 10) truncated at 0
  lpb <- dgamma(beta, shape = 0.01, rate = 0.01, log = TRUE) # Gamma(0.01, 0.01)
  lpg <- dexp(gamma, rate = 15, log = TRUE) # Exp(15)
  lpt <- -abs(theta) / 0.35 # Laplace(0, 0.35) truncated to (-1,1)
  lpa + lpb + lpg + lpt # Uniform(0,20) on sigma contributes constant
}

# Log posterior
log_post <- function(state, x, y, n) {
  alpha <- state[1]; beta <- state[2];
  gamma <- state[3]
  theta <- state[4]; sigma <- state[5]
  lp <- log_prior(alpha, beta, gamma, theta, sigma)
  if (!is.finite(lp)) return(-Inf)
  log_lik(alpha, beta, gamma, theta, sigma^2, x, y, n) + lp
}

# Single Gibbs sweep with three MH blocks; tracks acceptance counts
gibbs_step_tracked <- function(state, prop_sd, x, y, n, acc) {
  lp_c <- log_post(state, x, y, n)

  # Block 1: (alpha, beta) jointly
  prop <- state
  prop[1:2] <- state[1:2] + rnorm(2, 0, prop_sd[1:2])
  lp_p <- log_post(prop, x, y, n)
  if (log(runif(1)) < lp_p - lp_c) { state <- prop; lp_c <- lp_p; acc[1] <- acc[1] + 1 }

  # Block 2: gamma alone
  prop <- state
  prop[3] <- state[3] + rnorm(1, 0, prop_sd[3])
  lp_p <- log_post(prop, x, y, n)
  if (log(runif(1)) < lp_p - lp_c) { state <- prop; lp_c <- lp_p; acc[2] <- acc[2] + 1 }

  # Block 3: (theta, sigma) jointly
  prop <- state
  prop[4:5] <- state[4:5] + rnorm(2, 0, prop_sd[4:5])
  lp_p <- log_post(prop, x, y, n)
  if (log(runif(1)) < lp_p - lp_c) { state <- prop; acc[3] <- acc[3] + 1 }
```

```r
  list(state = state, acc = acc)
}

# Chain runner
run_chain_tracked <- function(start, n_iter, prop_sd, x, y, n) {
  mat <- matrix(NA, n_iter, 5,
                dimnames = list(NULL, c("alpha","beta","gamma","theta","sigma")))
  acc <- c(0, 0, 0)
  state <- start
  for (i in seq_len(n_iter)) {
    out <- gibbs_step_tracked(state, prop_sd, x, y, n, acc)
    state <- out$state
    acc <- out$acc
    mat[i,] <- state
  }
  list(chain = mat, accept_rates = acc / n_iter)
}

# Example: run chain from MLE starting values, 54000 iterations, discard first 4000
prop_sd <- c(alpha = 0.10, beta = 0.10, gamma = 0.03, theta = 0.25, sigma = 0.25)
burn <- 1000
n_iter_c <- burn + 50000

mle_sigma <- sqrt(mle["sigma2"])
start1 <- c(mle["alpha"], mle["beta"], mle["gamma"], mle["theta"], mle_sigma)

set.seed(42)
out <- run_chain_tracked(start1, n_iter_c, prop_sd, x, y, n)
post <- as.data.frame(out$chain[(burn + 1):n_iter_c, ])
post$sigma2 <- post$sigma^2
post$sigma <- NULL
```